Notes for Differential Geometry

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1 Local theory of curves

Definition 1.0.1. A curve $\vec{r} = \vec{r}(t)\colon (a,b) \to \mathbb{R}^n$ where $\vec{r}(t) = (x_1(t), \cdots, x_n(t))$ is **regular** if $x_i(t) \in C^\infty(a,b)$ and for all $t \in (a,b)$, we have $|\vec{r}\,'(t)| \neq 0$.

Specifically, we often take $n = 2, 3$.

We introduce the following notations. For a general variable like t, $\vec{r}\,' = \frac{\mathrm{d}}{\mathrm{d}t}\vec{r}(t)$.

Definition 1.0.2. The arc length of $\vec{r}$, denoted by s, is defined by

$$s = \int_a^t |\vec{r}\,'(\tau)|\,\mathrm{d}\tau.$$

Hence, we have $\frac{\mathrm{d}s}{\mathrm{d}t} = |\vec{r}\,'(t)|$. We use a different notation to denote the derivative of $\vec{r}$ with respect to s: $\dot{\vec{r}}(s) = \frac{\mathrm{d}}{\mathrm{d}s}\vec{r}(s)$. The benefit of using the arc length as the variable is the normalization of the tangent vector. That is,

$$\left|\dot{\vec{r}}(s)\right| = \frac{\mathrm{d}\vec{r}}{\mathrm{d}s} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}s} = \vec{r}\,' \cdot \frac{1}{|\vec{r}\,'|} = 1.$$

Let $T(s) = \dot{\vec{r}}(s)$ be the tangent vector and $N(s)$ be the normal vector with $|N| = 1$. We always assume that T is rotated counterclockwise by $\pi/2$ to obtain N.

1.1 Plane curves

Let $\vec{r} = \vec{r}(s)$, $T(s) = \dot{\vec{r}}(s)$ with $|T(s)| = 1$. Then $|T(s)|^2 = 1$. We have $\frac{\mathrm{d}}{\mathrm{d}s}|T(s)|^2 = 0$. i.e. $\frac{\mathrm{d}}{\mathrm{d}s}\langle T(s), T(s)\rangle = 2\langle T, \dot{T}\rangle = 0$.

2 Local theory of surfaces

3 Exterior differential systems